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Professional Summary

Computer Science graduate (GPA 3.83/4) and three-time national prize winner in physics and mathematics (2021), targeting a junior **Machine Learning Engineer** role. Hands-on experience in **deep learning**, **data analysis** and **real-time 3D simulation**: built and released a **3D solar eclipse simulator** (Unreal Engine 5, Python) with eclipse timings within **1.5 seconds** of the community's reference tool and an honorable mention at a student scientific session, and trained models reaching **97.2%** accuracy on satellite-image classification and an **R-squared of 0.94** on car-price prediction. Open to Data Scientist and Software Engineer positions, full-time or part-time.

Education

National University of Science and Technology Politehnica Bucharest

M.Sc. in Data Science (early admission, May 2026)

Sep 2026 – Present

B.Sc. in Computer Science, Cumulative GPA: 3.83/4

Oct 2022 – Jul 2026

Relevant Coursework: Data Structures and Algorithms, OOP, Parallel and Distributed Algorithms, Operating Systems, Programming Paradigms, Communication Protocols, Advanced Graphic Programming, Machine Learning.

Saint Sava National College

Sep 2018 – Jun 2022

Notable achievements:

- **1st Prize**, National Physics Contest “Phi” 2021
- **3rd Prize**, “Gazeta Matematică” National Olympiad 2021
- **3rd Prize**, National Physics Olympiad 2021

Projects

3D Solar Eclipse Simulator (Bachelor's Thesis) | Computer Graphics & Astronomy Jan 2026 – Jun 2026

- Developed a real-time simulator of all four solar eclipse types in **Unreal Engine 5** that shows the Moon's shadow (umbra and penumbra) as **3D cones** over realistic **Cesium** terrain, with the path of totality drawn as a 3D band on an accurate **WGS84** Earth model.
- **Unique value:** none of the 5 existing tools compared (Stellarium, Celestia, SpaceEngine, The Photographer's Ephemeris, Xavier Jubier's simulator) shows these shadow cones, and most offer only 2D maps or tables.
- Validated eclipse timings against Xavier Jubier's reference tool, achieving **1 s** accuracy for the 12 Aug 2026 eclipse and **1.5 s** for 2 Aug 2027, while Stellarium was off by 8 minutes for the 2017 eclipse.
- Designed a two-part system: Python calculates celestial positions (**Skyfield**, NASA JPL **DE441** data) and sends them to Unreal Engine over **UDP** every frame, achieving **60 FPS** by running slow calculations on a separate thread.
- Processed **71,264 eclipses** over **30,000 years** (13,000 BC – 17,000 AD) and **5,079 stars**, and identified **719 eclipses** total or annular over Romania.
- Added a catalogue browser, time controls, bookmarks, weather effects, a Romanian/English interface, and other planet-moon systems (Jupiter, Saturn, Neptune).
- **Tech stack:** Unreal Engine 5, C++, Python, Skyfield, Cesium for Unreal, JPL DE441, GeoPy, UDP, Multithreading.
- **Released** as a downloadable product on Gumroad: gabyboncan.gumroad.com/1/txlkbg

CannyCore – Parallel Canny Edge Detection (team of 3) | Parallel & GPU Computing Oct 2025 – Jan 2026

- Implemented the full **Canny edge-detection** pipeline in C++ and parallelized it in 8 variants (**Pthreads**, **OpenMP**, **MPI**, **MPI+OpenMP**, **MPI+Pthreads**, **CUDA**, **CUDA+OpenMP**, **CUDA+MPI**), reaching a **44.6× speedup** on 8K images with CUDA.
- Ran the full pipeline on the GPU (one thread per pixel, shared-memory Gaussian blur, hysteresis rewritten without recursion): 0.18 s vs. about 8 s serial on an 8K image; the best CPU variant, **MPI+Pthreads** (4 processes × 2 threads), reached **6.4×**, versus 3.4× for Pthreads and 3.0× for OpenMP.
- Profiled every variant with **Intel VTune**, **perf** and **Nsight Compute**: Gaussian blur took 48.6% of the serial runtime, while MPI_Bcast communication took 36.8% of the pure-MPI runtime; the results guided the design of the hybrid variants (halo rows, separable blur).
- Benchmarked all variants on 7 image resolutions (612×408 to 8000×8000) and several thread and process counts, and showed that CPU speedups saturate because of the sequential hysteresis step (Amdahl's law) and communication overhead.
- **Tech stack:** C++, CUDA, OpenMP, MPI, Pthreads, Intel VTune, Nsight Compute, perf, Make, Git.
- **Repository:** [Github](#)

Visual Classification & Linguistic Analysis Systems | Computer Vision & NLP Jan 2026

- Built and compared a **ResNet9 CNN** and a **HOG+MLP** for **image classification** in PyTorch on two 10-class datasets, reaching **97.2%** accuracy on satellite LandPatches (+26 percentage points over the MLP) and **87.0%** on

ImageBits (+17 points).

- Applied **transfer learning** (a model pre-trained on ImageBits, fine-tuned on only 2,000 LandPatches images), achieving **92.4%** accuracy, together with **Mixup** and **CoarseDropout** data augmentation.
- Developed **LSTM** and **RNN** models for Romanian sentiment analysis on **28,946 reviews** with **FastText** word embeddings, achieving **88.5%** accuracy (F1 score of about 0.90); the LSTM trained far more reliably than the RNN.
- Automated the experiment analysis (7 Python modules that produce EDA, **t-SNE**, WordCloud and training-log charts) and documented the results in a 140-page report.
- **Tech stack:** Python, PyTorch, Albumentations, FastText, AdamW, OneCycleLR, NumPy, scikit-learn, Matplotlib, pandas.
- **Repository:** [Github](#)

Predictive Regression Systems | Multi-Dataset Analytics

Nov 2025

- Developed and benchmarked **7 regression models** with **5-fold cross-validation** on used-car prices (**23,734 Autovit listings**) and bike-rental demand (**10,886 records**), achieving an **R-squared** of **0.94** and **0.90** with the best models.
- **Car prices:** polynomial **SVR** beat the linear baseline (R-squared 0.94 vs. 0.84), while Random Forest overfitted (0.95 on training data, 0.79 on test data).
- **Bike demand:** **Gradient Boosting** achieved MAE 22 versus 41 for Linear Regression.
- Added recent-history features (values from the previous day and week, rolling averages), raising Random Forest to an R-squared of 0.98 (MAE 6.8).
- Prepared the data with **EDA**, missing-value imputation, **Isolation Forest** outlier removal, log-transforms and 34 engineered features (e.g., km per year).
- **Tech stack:** Python, scikit-learn, pandas, NumPy, seaborn, joblib.
- **Repository:** [Github](#)

AI Medical Diagnosis Assistant | NLP & Semantic Search

Jun 2025 – Sep 2025

- Developed an educational prototype that turns free-text symptoms into a list of likely conditions using semantic search (**Sentence Transformers** embeddings, **K-Nearest Neighbors** with cosine similarity, **PCA** for speed), with model caching, an offline **TF-IDF** fallback and a desktop chatbot built with **Gradio** and **PyWebView**.
- **Tech stack:** Python, Sentence Transformers, scikit-learn, pandas, NumPy, joblib, Gradio, PyWebView.
- **Repository:** [Github](#)

AI Game Solver – Sokoban | Autonomous Search & Optimization

Mar 2025

- Developed a push-only Sokoban puzzle solver with **LRTA*** and obstacle-aware **A*** and **BFS** heuristics with deadlock detection, solving **8 of 9** benchmark maps in under 14 s each.
- LRTA* explored up to 13 times fewer states than **Simulated Annealing** (5,351 vs. over 70,000), and shortened solution paths by **48–87%** (69% on average); built a **Tkinter** viewer to replay each solution step by step.
- **Tech stack:** Python, Tkinter, A*, BFS, LRTA*, Simulated Annealing, Heuristic Search.
- **Repository:** [Github](#)

Skills Summary

- **Programming Languages:** Python, C++, C, SQL, Java, Kotlin, Bash, HTML
- **Machine Learning & Data:** PyTorch, scikit-learn, pandas, NumPy, Albumentations, CNN, RNN/LSTM, NLP, Computer Vision, Transfer Learning, Regression, Cross-Validation, Hyperparameter Tuning, EDA, Feature Engineering, FastText, Sentence Transformers, Semantic Search, PCA, TF-IDF
- **Computer Graphics & 3D:** Unreal Engine 5, Cesium for Unreal, 3D Simulation, Computer Graphics, Astronomical Computation (Skyfield, JPL DE441)
- **Software Engineering:** Data Structures & Algorithms, OOP, Multithreading, UDP Networking, Search Algorithms (A*, BFS, LRTA*, Simulated Annealing)
- **Tools & Platforms:** Git, GitHub, PyCharm, IntelliJ, Linux, Windows
- **Languages:** Romanian (native), English B2, French A2–B1

Activities & Awards

- **Presenter and Honorable Mention**, Scientific Communications Session for Students May 2026
 - Presented the bachelor's thesis project (3D Solar Eclipse Simulator), later released as a downloadable product.
- **Participant**, 3DPUB Summer School (Computer Graphics) Jun 2025 – Sep 2025
 - Developed the AI Medical Diagnosis Assistant (see Projects).

Certificates

- **ECDL Standard Certificate** (7 Modules)

2021